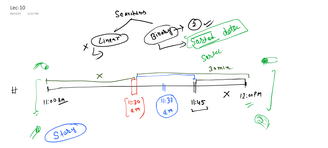
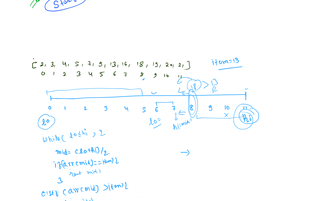
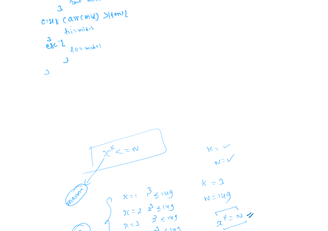
Lec 10
06/05/20 - 3:00 PM

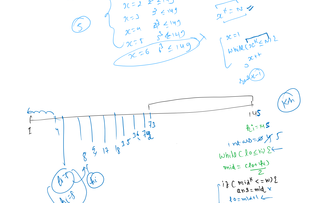


[2, 3, 4, 5, 2, 3, 3, 3, 14, 18, 13, 24, 2]

0 1 2 3 4 5 6 7 8 9 10



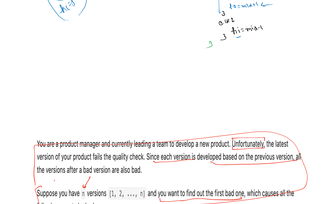


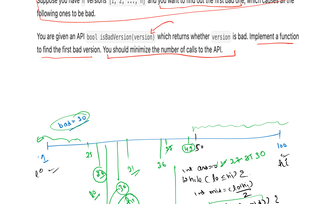


You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad.

Suppose you have n versions $[1, 2, \ldots, n]$ and you want to find out the first bad one, which causes all the following ones to be bad.

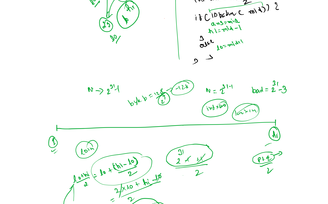
You are given an API `bool isBadVersion(version)` which returns whether `version` is bad. Implement a function to find the first bad version. You should minimize the number of calls to the API.





Farmer John has built a new long barn, with N (2 ≤ N ≤ 100,000) stalls. The stalls are located along a straight line at positions $x_1, \ldots, x_N$ (0 ≤ x_i ≤ 1,000,000,000).

His C (2 ≤ C ≤ N) cows don't like this barn layout and become aggressive towards each other once put into a stall. To prevent the cows from hurting each other, FJ wants to assign the cows to the stalls, such that the minimum distance between any two of them is as large as possible. What is the largest minimum distance?



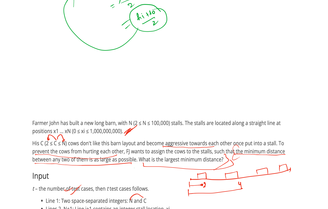
Input

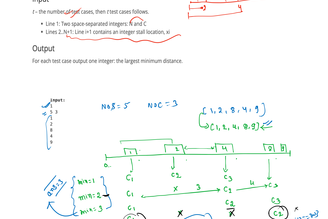
r - the number of test cases, then r test cases follows.

- Line 1: Two space-separated integers: N and C
- Lines 2..N+1: Line i+1 contains an integer stall location, x_i

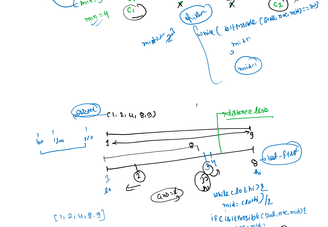
Output

For each test case output one integer: the largest minimum distance.

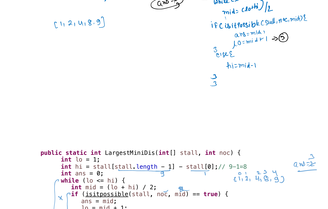





```
public static int LargestMiniDis(int[] stall, int noc) {
    int lo = 1;
    int hi = stall[stall.length - 1] - stall[0];// 9-1=8
    int ans = 0;
    while (lo <= hi) {
        int mid = (lo + hi) / 2;
        if (isPossible(stall, noc, mid) == true) {
            ans = mid;
            lo = mid + 1;
        } else {
            hi = mid - 1;
        }
    }
    return ans;
}
```



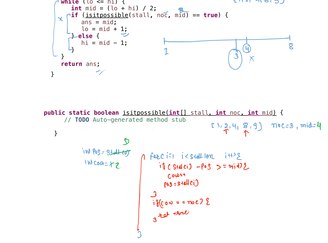

```
public static boolean isPossible(int[] stall, int noc, int mid) {
    // TODO Auto-generated method stub
}
```



Given an array `nums` with `n` objects colored red, white, or blue, sort them **in-place** so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers `0`, `1`, and `2` to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Example 1:

```
Input: nums = [2,0,2,1,1,0]
Output: [0,0,1,1,2,2]
```